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Cobalt-Catalyzed Chemo-, Regio-, Diastereo-, and Enantioselective Cascade Coupling of 1,1-Disubstituted Allenes and Aldehydes

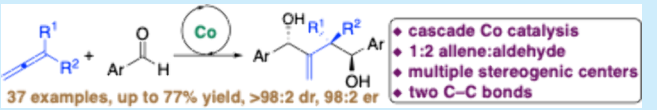
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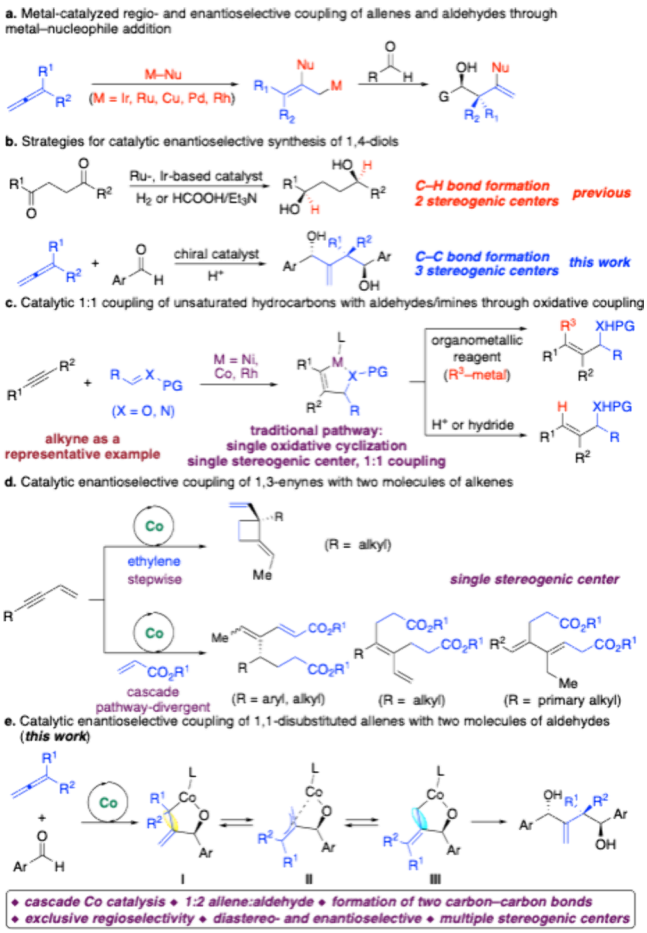
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ABSTRACT: A cobalt-catalyzed protocol for cascade coupling of 1,1-disubstituted allenes with aldehydes initiated by chemo-, regio-, diastereo-, and enantioselective oxidative cyclization of allenes and aldehydes followed by chemo-, regio-, and stereoselective addition of the resulting allyl−cobalt intermediates to the second molecule of aldehydes promoted by a chiral bisphosphine−cobalt complex is presented. Such processes represent an unprecedented reaction pathway for cobalt catalysis that enables simultaneous and precise incorporation two molecules of aldehydes onto the more sterically congested alkene of the allenes and construction of three stereogenic centers, delivering a range of enantioenriched 1,4-diols in up to 77% yield, >98:2 dr, and 98:2 er.



Scheme 1. Background and Reaction Design



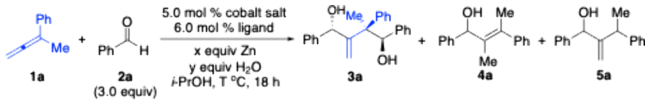
Transition metal-catalyzed enantioselective coupling of unsaturated hydrocarbons and aldehydes provides a straightforward and modular approach for the construction of chiral alcohols without the need for prefunctionalization.[1] Compared with reactions of unsaturated hydrocarbons with a single reactive site such as alkynes, multiple reactive sites of allenes complicated the chemo-, regio-, and stereoselectivity issues. Therefore, transformations of allenes have remained rare and more challenging. Significant progresses on catalytic enantioselective coupling of allenes and aldehydes promoted by Ir-based,[2] Ru-based,[3] Cu-based,[4] Pd-based,[5] and Rh-based[6] catalysts have been made (Scheme 1a). Such reactions proceeded through sequential nucleophile−metal addition to allenes followed by enantioselective reactions of the resulting allyl−metal species with aldehydes. Based on the reaction mechanism, only one molecule of aldehyde was able to be fused with one molecule of allene, enabling the formation of a single hydroxyl group. In contrast, the simultaneous incorporation of two molecules of electrophilic aldehydes onto allenes to generate enantioenriched diols in a cascade process remained unknown. Enantioenriched diols are important skeletons of various natural products, drugs, and chiral building blocks in organic synthesis. The development of efficient and stereoselective methods to construct enantioenriched diols is thus sought after in both academia and industry. Compared with multiple strategies for catalytic enantioselective synthesis of 1,2- and 1,3-diols, access to enantioenriched 1,4-diols

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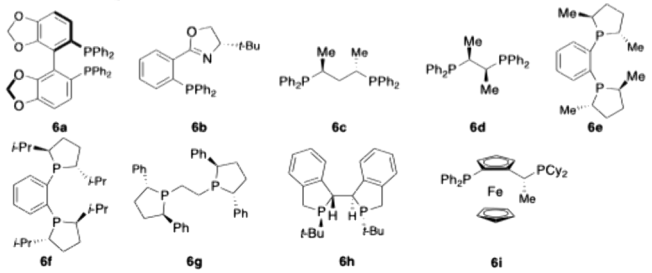


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Table 1. Optimization of the Reaction Conditions



entry	ligand	*x*	*y*	cobalt salt	*T* (°C)	yield of **3a** (%)[a]	yield of **4a** + **5a** (%)[b]	dr of **3a**[b]	er of **3a**[c]
1	**6a**	2.0	2.0	CoI_2	25	<5	93	NA[d]	NA[d]
2	**6b**	2.0	2.0	CoI_2	25	<5	<5	NA[d]	NA[d]
3	**6c**	2.0	2.0	CoI_2	25	<5	58	NA[d]	NA[d]
4	**6d**	2.0	2.0	CoI_2	25	<5	20	NA[d]	NA[d]
5	**6e**	2.0	2.0	CoI_2	25	<5	89	NA[d]	NA[d]
6	**6f**	2.0	2.0	CoI_2	25	25	57	68:32	78:22
7	**6g**	2.0	2.0	CoI_2	25	<5	46	NA[d]	NA[d]
8	**6h**	2.0	2.0	CoI_2	25	<5	66	NA[d]	NA[d]
9	**6i**	2.0	2.0	CoI_2	25	55	<5	88:12	95:5
10	**6i**	2.0	2.0	$CoBr_2$	25	63	<5	88:12	94:6
11	**6i**	2.0	2.0	$CoCl_2$	25	52	<5	87:13	92:8
12	**6i**	3.0	2.0	CoI_2	25	66	<5	88:12	96:4
13	**6i**	4.0	2.0	CoI_2	25	61	<5	87.5:12.5	96:4
14	**6i**	3.0	2.0	CoI_2	22	65	<5	89:11	97:3
15	**6i**	3.0	2.0	CoI_2	30	61	<5	85:15	96:4
16	**6i**	3.0	2.0	CoI_2	40	63	<5	76:24	95:5
17	**6i**	3.0	1.5	CoI_2	22	69	<5	89:11	97:3
18	**6i**	3.0	1.0	CoI_2	22	58	<5	89:11	97:3
19	**6i**	3.0	4.0	CoI_2	22	52	<5	80:20	92:8
20[e]	**6i**	3.0	1.5	CoI_2	22	72	<5	89:11	97:3



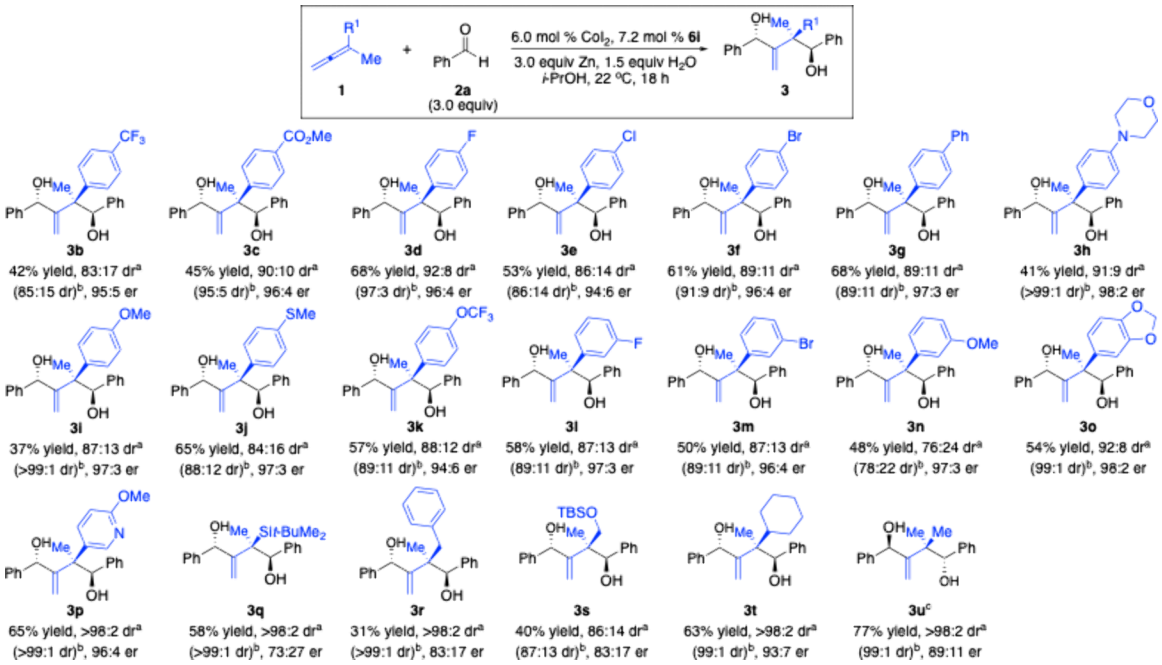
[a]Yield of the isolated product. [b]Determined by analysis of ^{1}H NMR spectra of the unpurified mixture. [c]Determined by analysis of HPLC spectra of the purified products. [d]Not available. [e]The reaction was performed in the presence of 6.0 mol % CoI_2 and 7.2 mol % **6i**.

containing remote stereogenic centers relied on catalytic enantioselective hydrogenation of 1,4-diketones, generating two stereogenic centers by C−H bond-forming reactions (Scheme 1b).[7] However, catalytic enantioselective preparation of 1,4-diols through carbon−carbon bond-forming reactions has remained elusive.

An alternative pathway for reactions of allenes and aldehydes is oxidative cyclization. Oxidative cyclization of two π-bonds promoted by low-valent transition metal complexes represents a classical elementary step in organometallic chemistry. Transition metal-catalyzed transformations through oxidative cyclization constitute one of the most important approaches for carbon−carbon bond-forming reactions in organic synthesis. Although protocols for catalytic enantioselective coupling of various unsaturated hydrocarbons and aldehydes/imines through oxidative cyclization have been reported, only one molecule of electrophilic aldehydes/imines could be incorporated (Scheme 1c).[8] In contrast to reactions of alkynes and 1,3-dienes, we developed the only example of cobalt-catalyzed divergent enantioselective 1:1 coupling of allenes and aldehydes through oxidative cyclization.[9] For 1:2 transformations of unsaturated hydrocarbons with electrophiles, pioneering work on cobalt-catalyzed enantioselective sequential coupling of 1,3-enynes with two molecules of ethylene was described by RajanBabu and co-workers (Scheme 1d).[10] Subsequently, our group developed a cobalt-catalyzed protocol for enantioselective divergent coupling of 1,3-enynes with two molecules of acrylates.[11] In both cases, two carbon−carbon bonds were formed at the alkyne moiety. Connecting two molecules of aldehydes to unsaturated hydrocarbons and incorporating two molecules of electrophiles onto an alkene moiety remained unknown. We imagined that taking advantage of our previous work on cobalt-catalyzed regiodivergent and enantioselective coupling of 1,1-disubstituted allenes and aldehydes,[9] the metallacycle (**I**, **II**, or **III**) containing an allyl−cobalt unit generated from oxidative cyclization of allenes and aldehydes might undergo allyl addition to the second molecule of aldehyde, enabling double functionalization of the more sterically congested alkene of the allenes (Scheme 1e). However, such a process is challenging. One of the key issue is chemoselectivity. The lower-lying LUMO of allenes led to a higher affinity for the metal center, and inherently, allenes have higher reactivity. Scarce examples of Rh-catalyzed and Co-catalyzed 2:1 coupling of allenes and aldehydes have been reported.[12] Moreover, an alcoholic solvent and water are required for catalyst turnover. Competitive protonation of the allyl−cobalt intermediate must be suppressed. In addition, a single chiral catalyst has to precisely control the regio- and

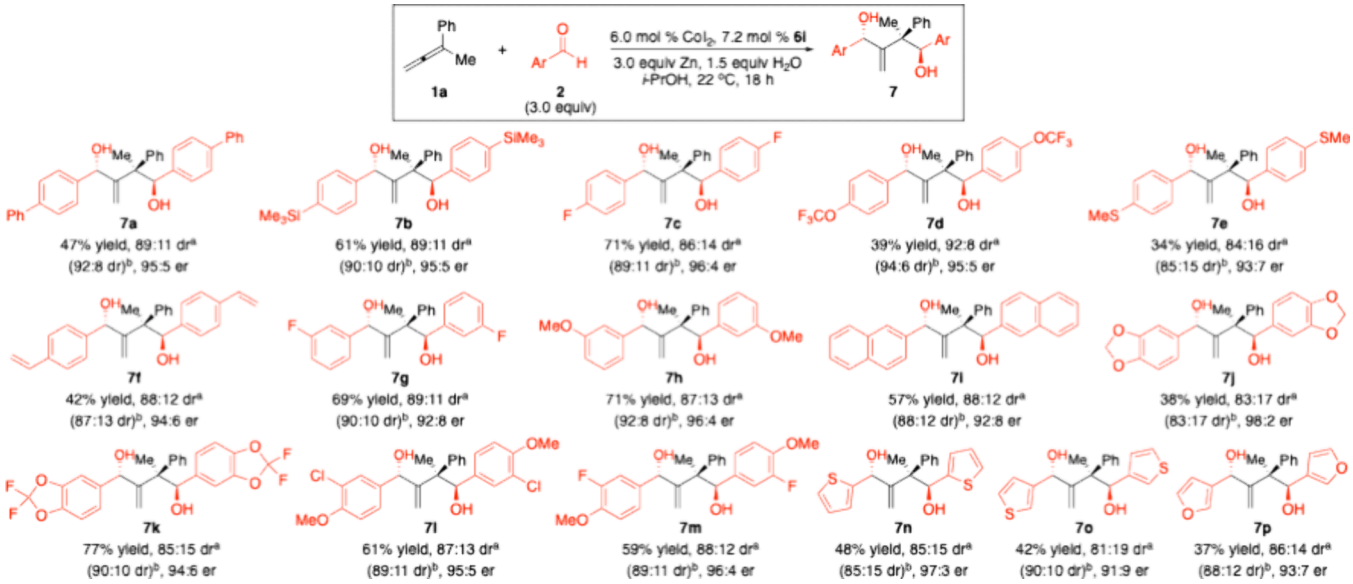
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Scheme 2. Scope of 1,1-Disubstituted Allenes



[a]Determined by analysis of ^{1}H NMR spectra of the unpurified mixture. [b]Determined by analysis of HPLC spectra of the isolated products. [c]The reaction was performed in the presence of 5.0 mol % CoI_2, 6.0 mol % **6f**, and 3.0 equiv of Zn in *i*-PrOH at 25 °C for 18 h.

Scheme 3. Scope of Aryl Aldehydes



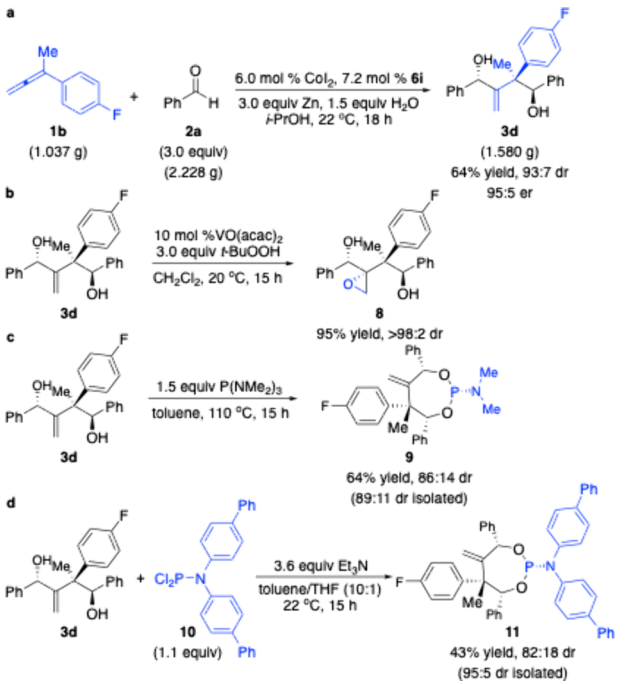
[a]Determined by analysis of ^{1}H NMR spectra of the unpurified mixture. [b]Determined by analysis of HPLC spectra of the isolated products.

stereoselectivity of the oxidative cyclization and aldehyde addition processes. Herein, we present an unprecedented protocol for chemo-, regio-, and enantioselective cascade reductive coupling of 1,1-disubstituted allenes with two molecules of aldehydes to afford a broad scope of 1,4-diols containing three stereogenic centers that are otherwise difficult to access.

We commenced our studies with the reaction of 1,1-disubstituted allene **1a** with benzaldehyde **2a** in the presence of various Co complexes (Table 1). Zn powder was required as the stoichiometric reductant. Axially stereogenic bisphosphine **6a**, which was the optimal ligand for enantioselective 1:1 reductive coupling of 1,1-disubstituted allene **1a** and benzaldehyde **2a** to afford chiral allylic alcohol **4a**, provided only 1:1 coupling products (**4a** and **5a**) in our previous work (entry 1).[9] Similarly, bisphosphines containing stereogenic centers at the alkyl backbone (**6c** and **6d**), chiral phospholane fragments (**6e** and **6g**), and P-stereogenic centers (**6h**) cannot promote the 1:2 coupling reaction (entries 3−5, 7, and 8, respectively). No reaction occurred with phosphinooxazoline ligand **6b** (entry 2). 1,4-Diol **3a** was obtained in 25% yield with 68:32 dr and 78:22 er for the major diastereomer in the

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Scheme 4. Gram Scale Reaction and Functionalization

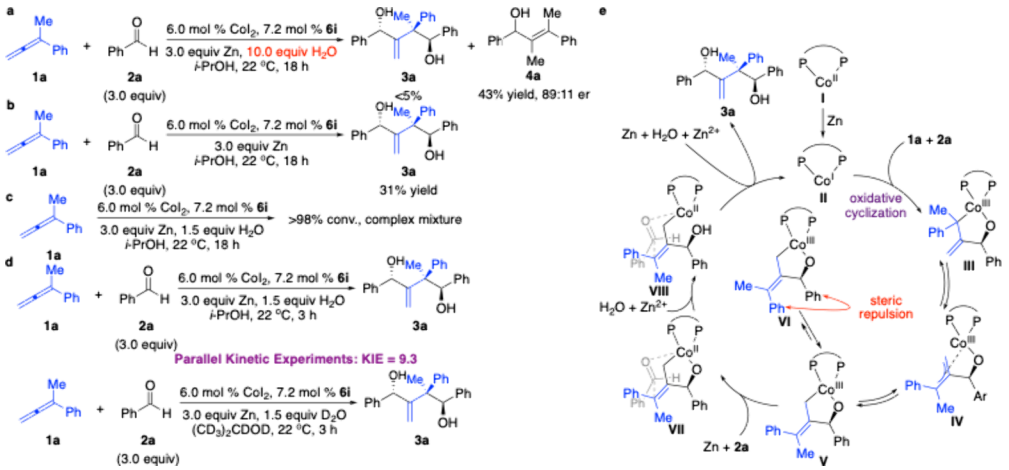


presence of optimal ligand **6f** for enantioselective 1:1 reductive coupling of disubstituted allene **1a** and benzaldehyde **2a** to afford allylic alcohol **5a** in our previous study (entry 6).[9] Cascade reaction of 1,1-disubstituted allene **1a** and benzaldehyde **2a** promoted by the Co complex derived from bisphosphine **6i** bearing a ferrocene skeleton delivered 1,4-diol **3a** exclusively in 55% yield with 88:12 dr and 95:5 er for the major diastereomer, enabling simultaneous establishment of three stereogenic centers (entry 9). Optimization of cobalt salts revealed that only cobalt halides could promote the reaction, albeit with erosion of enantioselectivity (entries 10 and 11). Increasing the amount of Zn led to an improvement in efficiency (entries 12 and 13). An increased reaction temperature resulted in a decrease in diastereoselectivity, whereas performing the reaction at 22 °C furnished 1,4-diol **3a** in 65% yield with 89:11 dr and 97:3 er for the major diastereomer (entries 14−16). Alteration of the amount of water suggested that the highest efficiency was obtained with 1.5 equiv of water (entries 17−19). A slight increase in the catalyst loading to 6.0 mol % CoI_2 and 7.2 mol % **6i** improved the efficiency, whereas further increasing the amount of the catalyst did not provide a better yield (entry 20).

With the optimal conditions in hand, we investigated the scope of allenes (Scheme 2). 1,1-Disubstituted allenes bearing electron-deficient (**3b−d** and **3l**), electron-rich (**3g−k**, **3n**, and **3o**), halogenated (**3d−f**, **3l**, and **3m**), and sterically congested (**3g** and **3o**) aryl groups underwent the cascade 1:2 reductive coupling, affording the enantioenriched 1,4-diols in 37−68% yields with 76:24−92:8 dr and 94:6−98:2 er for the major diastereomer. Pyridyl-substituted allene was well tolerated (**3p**). Functional groups such as ester (**3c**) and amine (**3h**) are compatible with the reaction conditions. Reaction of silyl-substituted allene delivered 1,4-diol **3q** in 58% yield with >98:2 dr albeit 73:27 er for the major diastereomer. Allenes substituted with two alkyl groups could be transformed into the desired 1,4-diols (**3r−t**) in 31−63% yields with 86:14 to >98:2 dr and 83:17−93:7 er for the major diastereomer. Bisphosphine **6f** was optimal for the transformation of commercially available dimethyl-substituted allene without the need for the addition of water, furnishing 1,4-diol **3u** in 77% yield with >98:2 dr and 89:11 er. It should be mentioned that in some cases (**3c**, **3d**, **3h**, **3i**, and **3o**), diastereomers could be separated by silica gel chromatography. Monosubstituted allenes could not be converted into the desired products. Oligomerization of allenes occurred preferentially.

We next explored the scope of the aldehydes (Scheme 3). Only aryl aldehydes could participate in the cascade reaction. Aliphatic aldehydes were unreactive under the reaction conditions. A variety of aldehydes containing electron-rich (**7a**, **7d**, **7e**, and **7h−m**), electron-deficient (**7c** and **7g**), halogenated (**7c**, **7g**, **7l**, and **7m**), and sterically congested (**7a** and **7i−m**) aryl groups could be converted into the desired 1,4-diols in 34−77% yields with 83:17−92:8 dr and 92:8−98:2 er for the major diastereomer. Functional groups such as silyl (**7b**), thiol ether (**7e**), and styrene (**7f**) are quite compatible. Although sizable 3,4-disubstituted benzaldehydes were able to be transformed into the desired 1,4-diols (**7j−m**), aryl aldehydes bearing a substituent at position 2 were unreactive. Reactions of heteroaryl aldehydes (**7n** and **7o**) delivered the enantioenriched 1,4-diols in 37−48% yields with 81:19−86:14 dr and 91:9−97:3 er for the major diastereomer.

Scheme 5. Mechanistic Studies



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The cascade reaction can be performed on a gram scale (Scheme 4a). Transformation of 1,1-disubstituted allene **1b** (1.037 g) with benzaldehyde **2a** (2.228 g) promoted by the Co complex derived from **6i** furnished 1,4-diol **3d** (1.580 g) in 64% yield with 93:7 dr and 95:5 er. Directed epoxidation of the 1,1-disubstituted alkene moiety induced by $VO(acac)_2$ enabled highly diastereoselective construction of other stereogenic centers in 95% yields (Scheme 4b).[13] Cyclization of 1,4-diol **3d** afforded P-stereogenic phosphoramidites **9** and **11** in 64% and 43% yields with 86:14 and 82:18 dr, respectively (Scheme 4c,d).[14,15] In particular, phosphoramidite **11** could be isolated with 95:5 dr, showing its potential as a new class of P-stereogenic phosphoramidite ligands.

To gain some mechanistic insight, we conducted a series of experiments (Scheme 5). Increasing the amount of water to 10 equiv led to complete suppression of the cascade 1:2 coupling reaction (Scheme 5a). Competitive 1:1 coupling product **4a** was formed exclusively. Such a result indicated that **3a** and **4a** were generated via the same metallacycle intermediate. The stereochemical control of the oxidative cyclization was not exclusive, causing diastereomers after the second aldehyde addition. Removal of water from the reaction conditions resulted in a significant decrease in the reaction efficiency, suggesting that water was crucial for the second aldehyde addition and/or catalyst turnover (Scheme 5b). Transformation of allene **1a** in the absence of aldehyde **2a** provided a complex mixture of self-oligomerization products, implying that the catalyst accurately controlled the chemoselectivity of the initial oxidative cyclization process (Scheme 5c). Parallel isotope kinetic experiments illustrated that cleavage of the O−H bond of water might be the rate-determining step (Scheme 5d).

Based on all of the experimental observations and literature precedents, we propose the possible catalytic cycle (Scheme 5e). Oxidative cyclization of allene **1a** and benzaldehyde **2a** promoted by Co(I) complex **II** in situ generated from Co(II) precursor **I** afforded Co(III) metallacycle **III** bearing an allyl−Co(III) moiety, which was able to undergo allyl isomerization via π-allyl−Co(III) intermediate **IV** to afford more stable allyl−Co(III) complex **V** containing a tetrasubstituted alkene. The 1,3-allylic strain in **VI** led to a more favorable formation of intermediate **V**. One-electron reduction of Co(III) metallacycle **V** to Co(II) species **VII** followed by allyl addition to benzaldehyde **2a** and subsequent protonation/exchange with a Zn salt released product **3a**. The second allyl addition might be accelerated by detaching the OH group and assisted by Zn salt and water (**VII** → **VIII**). Reduction of the Co(II) species with Zn regenerated the catalyst.

In conclusion, we have disclosed an unprecedented protocol for chemo-, regio-, diastereo-, and enantioselective cascade coupling of 1,1-disubstituted allenes and aldehydes promoted by a Co complex derived from a commercially available bisphosphine ligand, affording a broad scope of enantioenriched 1,4-diols that are otherwise difficult to access and enabling simultaneous construction of three stereogenic centers. The synthetic utility was demonstrated by functionalization of the products, delivering densely functionalized 1,4-diols and P-stereogenic phosphoramidites. Mechanistic studies revealed a novel cascade reaction pathway for cobalt catalysis. Such discoveries unveiled new opportunities for designing new cobalt-catalyzed enantioselective reactions and pushing forward the development of organocobalt chemistry. Further investigations of other cobalt-catalyzed enantioselective cascade transformations are underway.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c03902.

Experimental details and full spectroscopic data for all new compounds (PDF)

Accession Codes

Deposition Numbers 2391013, 2425175, 2441131, and 2441142 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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